

Kodbud Python Internship

Code Submission — Part 2 of 2

Name: Raut Avishkar H.

Task 5 — Password Strength Checker

Validates a password against five criteria using Python's re (regex) module and string methods. Gives a score-based rating — Weak, Moderate, or Strong — with actionable feedback for each failed rule.

■ Source Code

```
# Task 5: Password Strength Checker
# Concepts: re module, string methods, conditional logic, f-strings

import re

def check_password(password: str) -> dict:
    rules = {
        "min_length" : len(password) >= 8,
        "has_upper" : bool(re.search(r"[A-Z]", password)),
        "has_lower" : bool(re.search(r"[a-z]", password)),
        "has_digit" : bool(re.search(r"\d", password)),
        "has_special" : bool(re.search(r'[@#$%^&*(),.?":{}|<>\_-=+\[\]\\'',
password)),
    }
    score = sum(rules.values())
    if score <= 2:
        rating = "Weak"
    elif score <= 4:
        rating = "Moderate"
    else:
        rating = "Strong"
    return {"rules": rules, "score": score, "rating": rating}

def display_result(password: str):
    result = check_password(password)
    rules, score, rating = result["rules"], result["score"], result["rating"]
    labels = {
        "min_length" : "Minimum 8 characters",
        "has_upper" : "At least one uppercase letter",
        "has_lower" : "At least one lowercase letter",
        "has_digit" : "At least one digit (0-9)",
        "has_special" : "At least one special character",
    }
    print("\n Password Analysis:")
    print(" " + "-" * 36)
    for key, label in labels.items():
        status = "PASS" if rules[key] else "FAIL"
        mark = "[+]" if rules[key] else "[-]"
        print(f" {mark} {label:<34} {status}")
    print(" " + "-" * 36)
```

```

print(f" Score : {score}/5")
print(f" Rating : {rating}")
if rating == "Weak":
    print(" Tip : Add uppercase, digits, and special characters.")
elif rating == "Moderate":
    print(" Tip : Almost there – fix the remaining rules above.")
else:
    print(" Your password is secure!")
print()

# Entry point
print("=" * 45)
print(" Password Strength Checker")
print("=" * 45)
while True:
    pwd = input("Enter a password to check (or 'quit'): ").strip()
    if pwd.lower() == "quit":
        print("Exiting checker.")
        break
    if not pwd:
        print(" Password cannot be empty.\n")
        continue
    display_result(pwd)

```

■ Sample Run

```

Enter a password to check: Hello@123

Password Analysis:
-----
[+] Minimum 8 characters PASS
[+] At least one uppercase letter PASS
[+] At least one lowercase letter PASS
[+] At least one digit (0-9) PASS
[+] At least one special character PASS
-----

Score : 5/5
Rating : Strong
Your password is secure!

```

Task 8 — Simple Chatbot (CLI)

A rule-based chatbot that matches user input via keyword detection and if-elif-else logic. Handles greetings, FAQs, small talk, and an unknown-input fallback. Includes a quit command to exit cleanly.

■ Source Code

```
# Task 8: Simple Chatbot (CLI)
# Concepts: if-elif-else, string methods, keyword matching, loops

def get_response(user_input: str) -> str:
    msg = user_input.lower().strip()

    if any(w in msg for w in ["hello", "hi", "hey", "howdy", "greetings"]):
        return "Hello! I'm PyBot. How can I help you today?"
    elif any(w in msg for w in ["bye", "goodbye", "see you", "exit", "quit"]):
        return "Goodbye! Have a great day!"
    elif any(p in msg for p in ["how are you", "how r you", "what's up"]):
        return "I'm just a bot, but I'm running perfectly! Thanks for asking."
    elif any(p in msg for p in ["your name", "who are you", "what are you called"]):
        return "I'm PyBot, a simple Python chatbot built for the Kodbud internship!"
    elif any(w in msg for w in ["python", "programming", "code", "coding"]):
        return ("Python is awesome! It's great for automation, web dev, "
                "data science, and AI. What would you like to know?")
    elif any(w in msg for w in ["internship", "task", "kodbud", "project"]):
        return ("This chatbot is part of a Kodbud Python internship. "
                "Complete 4 out of 8 tasks and upload them to LinkedIn!")
    elif any(w in msg for w in ["time", "date", "today", "day"]):
        from datetime import datetime
        now = datetime.now()
        return f"Current date & time: {now.strftime('%A, %d %B %Y - %I:%M %p')}}"
    elif any(w in msg for w in ["joke", "funny", "laugh", "humor"]):
        return ("Why do programmers prefer dark mode?\n"
                "Because light attracts bugs! :-)")
    elif any(w in msg for w in ["thank", "thanks", "thx", "appreciate"]):
        return "You're welcome! Feel free to ask me anything else."
    elif "help" in msg or "?" == msg:
        return ("I can chat about:\n"
                "- Greetings & small talk\n"
                "- Python programming\n"
                "- Kodbud internship info\n"
                "- Current date/time\n"
                "- Jokes!\n"
                "Just type naturally and I'll do my best.")
    else:
        return ("Hmm, I didn't quite understand that. "
                "Try asking about Python, the internship, or type 'help'.")

def run_chatbot():
    print("=" * 50)
    print(" PyBot – Simple Python Chatbot")
    print("=" * 50)
    print("Type 'quit' or 'bye' to exit.\n")
    while True:
        user_input = input("You : ").strip()
```

```
if not user_input:
    continue
response = get_response(user_input)
print(f"PyBot : {response}\n")
if any(w in user_input.lower() for w in ["quit", "bye", "goodbye", "exit"]):
    break

# Entry point
run_chatbot()
```

■ Sample Run

```
You : Hello
PyBot : Hello! I'm PyBot. How can I help you today?

You : Tell me a joke
PyBot : Why do programmers prefer dark mode?
Because light attracts bugs! :-)

You : What is Python?
PyBot : Python is awesome! It's great for automation, web dev,
data science, and AI. What would you like to know?

You : bye
PyBot : Goodbye! Have a great day!
```